

# Adam Turaj

+1 (613) 880-8614 | aturaj@uwaterloo.ca | github.com/AdamTuraj | adamturaj.com

## EDUCATION

### University of Waterloo

BASc, Mechatronics Engineering (Co-op) **GPA: 3.7/4.0**

Waterloo, ON

Sep 2025 – Apr 2030 (Expected)

## TECHNICAL SKILLS

**Electrical:** Altium, KiCad, PCB design, FEMM, sensor integration, soldering

**Programming:** Python, C++, SystemVerilog, Bash

**CAD / Design:** SolidWorks, Fusion 360, 3D printing, technical drawings

## EXPERIENCE

### Zipline

South San Francisco, CA

Electrical Intern

Sep 2026 – Present

- Evaluate electrical technologies with the New Products team to reduce product weight and complexity.
- Design an inductive wireless power and data system, including coil, driver and rectifier, using **FSK/ASK** for bidirectional, full-duplex communication.
- Modeled the coil in **FEMM** and optimized its geometry in **Python** to reduce mass and losses.

### Formula Electric (Formula SAE)

University of Waterloo

Electrical Team Member

Sep 2025 – Present

- Design a **28-channel** battery simulator to test voltage-sensing boards: **3–4.2 V**, **1 mV** accuracy and **250 mA** four-quadrant output.
- Designed a **4-layer Altium battery breakout PCB** for a high-voltage pack, integrating sensor interfaces, control lines and **15 A** fan power distribution.
- Developed a bidirectional **915 MHz** headset radio PCB targeting **2 km** range, with SAI and an audio codec.
- Wired and calibrated the data acquisition system; helped route the next wire harness and integrated accelerometer and pressure sensors with **CAN** and **MoTeC** telemetry.

### Ciena

Ottawa, ON

FPGA Verification Co-op

Jan 2026 – Apr 2026

- Built a reusable **SystemVerilog** testbench for a production core across **4 projects**; used internal signals to validate Avalon write captures around an IP register-model limitation.
- Resolved a **1-in-16 regression failure** through waveform analysis and root-cause isolation; validated the fix with a senior engineer.
- Added **4 configurable options** to **Perl** scripts to streamline verification environment generation.

## PROJECTS

### BLDC ESC — Sensored Motor Driver

Aug 2025 – Sep 2025

- Designed a **25 A**, **8.4 V** sensored BLDC motor driver in **KiCad** for **2S LiPo** applications.
- Routed a **4-layer PCB** with high-current paths and **DRV8323** single-PWM control to simplify firmware.

### ESP-CAM — Single-Stack Camera Board

Mar 2026 – Apr 2026

- Designed a **35 mm × 41 mm ESP32-CAM PCB** for lightweight drone vision.
- Prioritized **OV2640** camera-interface and **8-bit data-bus** routing for reliable image capture.

### Hackboard — STM32 Development Board

Aug 2024

- Designed an **STM32L476RG** board with UNO-shield support, **45 GPIO pins** and microSD.
- Combined a high-power MOSFET output with low-power design for up to **150 hours** on a **9 V** battery.